

PICBOY

INSTRUCTION BOOKLET

Handheld Gaming Console - Group (5)

Powered by Raspberry Pi Pico W (RP2040)



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SYSTEM OVERVIEW

The PicoBoy is a handheld gaming console built around the Raspberry Pi Pico W microcontroller. It supports two original games and uses a real-time motion sensor for tilt-based controls.

The PicoBoy includes:

- 2.4-inch colour TFT LCD display (240 × 320 pixels)
- 8 tactile push-button controls
- LSM6DSL six-axis IMU for motion sensing
- Mono speaker with LM386 amplifier
- 3.7V Li-Po battery with USB-C charging
- 3D-printed PLA enclosure

DESCRIPTION OF COMPONENTS

The front exterior view of the PicoBoy is shown in Fig. 1. This view identifies the main physical features referenced throughout this booklet.

1. 2.4" TFT LCD Screen: Displays all game graphics in colour at 240 × 320 pixels. Driven by the ILI9341 controller via SPI.
2. D-Pad (↑ ↓ ← →): Four-way directional pad used for character movement and menu navigation.
3. A Button: Primary action button used to jump, confirm, or interact depending on the game.
4. B Button: Secondary action button used to cancel or trigger alternate actions.
5. START Button: Used to start, confirm menu selections, or restart depending on the game mode.
6. SELECT Button: Opens pause options or returns to the previous menu where applicable.

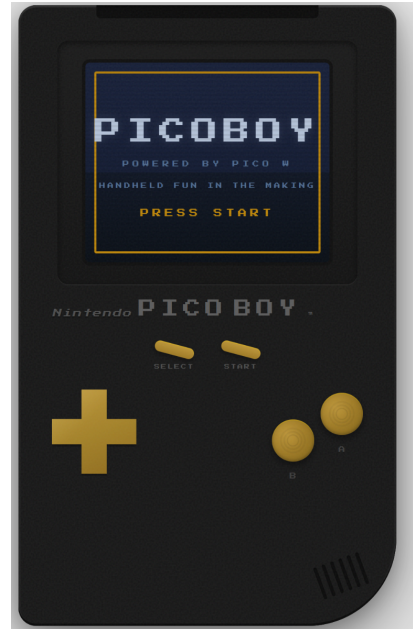


Figure 1: Front view of the PicoBoy

7. Speaker Grille: Mono audio output. Do not block during use.
8. Power Switch: Slide to ON to power the system. Slide back to OFF to power down.
9. USB Type-C Port: Used for charging the battery.
10. Micro-USB Port: Used to update firmware and/or power the PicoBoy directly.
11. BOOTSEL Access: Recessed button on the back of the case. Hold while connecting the Micro-USB cable to update firmware.
12. SD Card Slot: Reserved for future use. Allows additional game and asset storage.
13. IMU Sensor (internal): LSM6DSL six-axis accelerometer and gyroscope. Allows tilt-based controls.
14. Li-Po Battery (internal): 3.7V, 2000 mAh rechargeable lithium-polymer battery. Charged via the USB Type-C port.

POWERING ON THE PICOBOY

Before First Use - Charge the Battery

1. Connect a USB-C cable to the charging port located at the bottom of the PicoBoy.
2. Allow approximately 2.5 to 3 hours for a full charge under normal conditions.
3. Once sufficient charging time has elapsed, disconnect the cable.

Alternatively, the PicoBoy can be powered and used while connected via the Micro-USB port, which supplies power directly to the internal microcontroller.

The device is shipped with the battery pre-charged, so charging may not be required before first use.

Switching On

1. Slide the power switch to the ON position.
2. The LCD screen turns on. The PicoBoy boot screen appears, as shown in Fig. 2.
3. Press START to proceed to the Profile selection screen.

Note: If the screen remains blank, make sure the battery is charged



Figure 2: PicoBoy boot screen

Switching Off

Slide the power switch back to the OFF position at any time.

PROFILES

After the boot screen, the Profile selection screen appears. Up to four player profiles can be stored. Each profile tracks its own coin total and high scores independently.

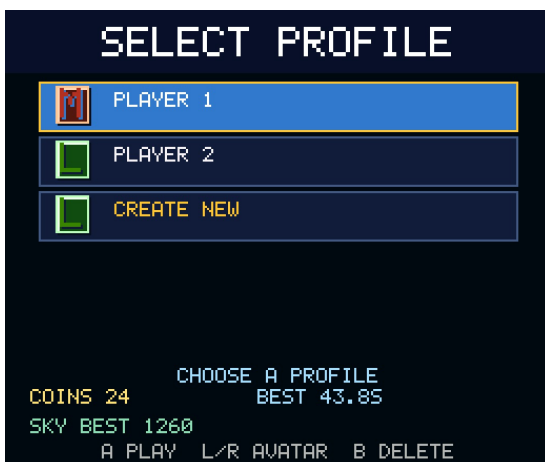


Figure 3: Selecting an existing profile

Creating a New Profile

1. Use $\uparrow$ / $\downarrow$ to highlight CREATE NEW.
2. Press $\leftarrow$ / $\rightarrow$ to choose an avatar icon.

3. Press A or START to confirm and create the profile.



Figure 4: Creating a new profile

Selecting a Profile

- Use $\uparrow$ / $\downarrow$ to highlight PLAYER 1 or PLAYER 2 on the profile screen shown in Fig. 3.
- Press A to load the profile and proceed to the Game Select screen.
- Press B to delete the highlighted profile.

Note: Profile statistics displayed on this screen include COINS collected, BEST time in Plumber Man, and BEST distance in Sky Dodger.

GAME SELECT SCREEN

After selecting a profile, the Game Select screen appears. It shows the active player's avatar, coin count, and best scores. Three options are available: Plumber Man, Sky Dodger, and Settings.



Figure 5: Game Select screen

1. Use $\uparrow$ / $\downarrow$ to highlight a game or Settings.
2. Press A or START to launch the selected option.
3. Press B or SELECT at any time to return to the Profile screen.

PLUMBER MAN - CONTROLS AND GAMEPLAY

Plumber Man is a side-scrolling platformer. Run and jump through levels, collect coins, and avoid enemies. Reach the flag pole at the end to progress to the next stage. Your fastest time is saved to your profile.



Figure 6: Plumber Man gameplay

Controls

Button	Action
← / →	Move left or right
A Button	Jump
SELECT	Pause the game and open the Pause Menu
START	Restart the game

Pause Menu

Press SELECT during gameplay to pause. The Pause menu allows volume adjustment and provides the option to return to the Game Select screen.



Figure 7: Pause menu with volume control

While paused:

- Press ↑ / → to increase volume. Press ↓ / ← to decrease volume.

Game Over

If your character collides with an enemy or falls into a gap, the game ends. The Game Over screen offers two options, shown in Fig. 8.

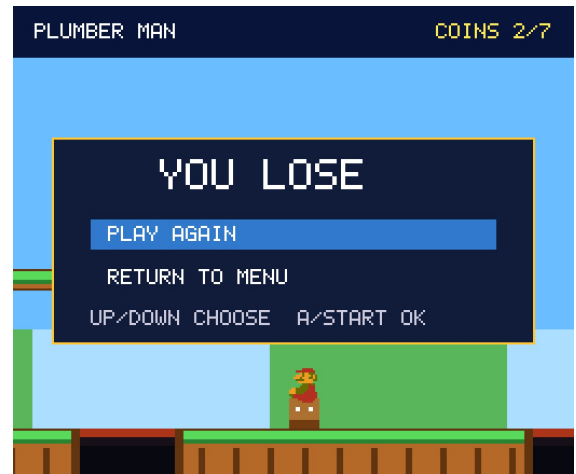


Figure 8: Game Over screen

- Use ↑ / ↓ to highlight PLAY AGAIN or RETURN TO MENU.
- Press A or START to confirm your choice.

Note: Coins collected during a failed run are not saved to your profile. Complete the level to save your coins and record your time.

SKY DODGER - CONTROLS AND GAMEPLAY

Sky Dodger is a tilt-controlled falling game. Your character descends from the sky while avoiding birds, barriers, and other airborne obstacles approaching from below and the edges. Collect coins along the way and survive as long as possible. Distance fallen and coins collected are tracked and saved to your profile.

IMU Calibration - Starting the Game



Figure 9: IMU calibration step 1

Before each run, Sky Dodger performs a two-step IMU calibration to establish your neutral and tilt positions.

1. **HOLD IT LEVEL** - Hold the PicoBoy vertically and completely level. Press A or START to capture the neutral position.
2. **TILT TO ONE SIDE** - Tilt the console to either side as far as comfortable. Press A or START to capture your tilt range.



Figure 10: IMU calibration step 2

Selecting Difficulty

Next, the difficulty selection screen appears. Higher difficulty levels increase obstacle speed and frequency.



Figure 11: Difficulty selection screen

1. Press $\uparrow$ / $\downarrow$ to change the difficulty level.
2. Press A or START to begin the game.

Controls

Control	Action
Tilt Left	Move character left
Tilt Right	Move character right
SELECT	Pause the game

MOTION CONTROLS (IMU)

The PicoBoy contains an LSM6DSL six-axis Inertial Measurement Unit (IMU) connected via I2C. This sensor measures the physical tilt of the console and converts the accelerometer readings into directional control signals.

Note: Hold the PicoBoy upright and steady for accurate tilt detection. Avoid rapid shaking or impacts during gameplay.

SETTINGS

The Settings screen is accessible from the Game Select screen. It allows adjustment of display mode, master volume, and audio output. A built-in Noise Diagnostic tool is also provided.

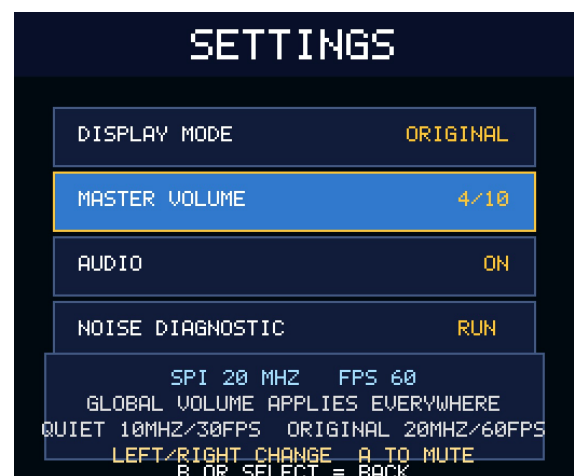


Figure 12: Settings screen

Settings Options

- **Display Mode** - Toggle between ORIGINAL (20 MHz SPI / 60 FPS) and QUIET (10 MHz SPI / 30 FPS). QUIET mode reduces bus activity and may lower audible electronic interference.
- **Master Volume** - Adjustable from 0 to 10. Press $\leftarrow$ / $\rightarrow$ to change. Press A to mute. Volume applies globally across all games.
- **Audio** - Toggle audio ON or OFF.
- **Noise Diagnostic** - Helps identify sources of audible electronic noise. Toggle SCREEN LOAD, BACK-LIGHT, and IMU POLL individually and listen after each change. Live IMU raw data (X, Y, Z axis) is displayed. Press START to exit.

Press B or SELECT to return to the Game Select screen.

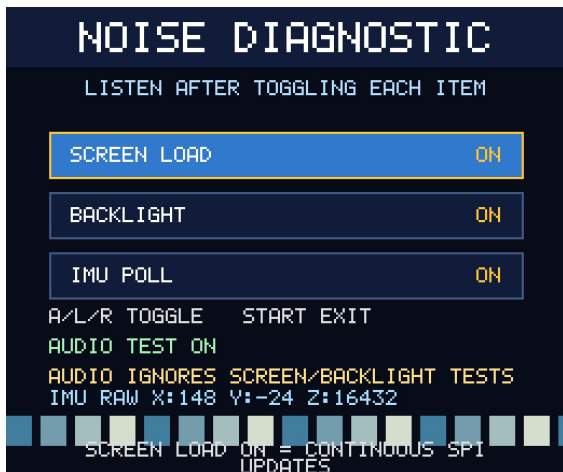


Figure 13: Noise Diagnostic tool

AUDIO OUTPUT

The PicoBoy generates audio using PWM from the Pico. The signal passes through a low-pass filter and is amplified by an LM386 amplifier before reaching the speaker.

Master volume can be adjusted in the Settings screen or from the Pause menu during gameplay.

Note: If electronic noise is audible from the speaker, use the Noise Diagnostic tool shown in Fig. 13 to help isolate the source.

REPROGRAMMING / FIRMWARE UPDATE

The PicoBoy can be reprogrammed via the Micro-USB port and BOOTSEL button. The BOOTSEL access opening is located on the back of the enclosure. When the unit enters bootloader mode, it appears as a USB mass storage drive.

1. Power off the PicoBoy.
2. Press and hold the BOOTSEL button - accessible through the opening on the back of the case.
3. While holding BOOTSEL, connect a Micro-USB from the port at the top of the PicoBoy to a computer.
4. The PicoBoy will appear as a removable drive on your computer. Release the BOOTSEL button.
5. Drag and drop the new firmware .uf2 file onto the drive.
6. The device will reboot automatically once the file has been moved.

Note: Do not disconnect the USB cable while the firmware file is being copied.

TROUBLESHOOTING

Before seeking assistance, try the following:

Problem	Solution
Screen remains blank when switched on.	Switch off, wait a moment, then switch back on. Connect USB-C and charge before retrying. If still blank, enter bootloader mode and reflash the firmware.
Buttons are unresponsive.	Switch off and back on. Check that no button is physically stuck. If the problem continues, reflash the firmware via USB.
Motion controls not working in Sky Dodger.	Repeat the calibration process shown on the IMU screens. Hold the console level during the neutral capture step. If the issue continues, restart the device and retry.
Audible electronic noise from speaker.	Open Settings and run the Noise Diagnostic tool. Toggle SCREEN LOAD, BACKLIGHT, and IMU POLL individually to identify the source. Switch to QUIET display mode in Settings to reduce SPI bus noise.
No audio from the speaker.	Check that Master Volume is not set to 0 in Settings. Ensure Audio is set to ON in Settings.
Device freezes during gameplay.	Switch off, wait five seconds, then switch on again. If freezes occur frequently, reflash the firmware via USB.
Device does not appear to charge.	Check that the USB-C cable is fully seated. Try a different USB power source rated at 5V. Allow enough time for charging, as the charging module is internal and has no visible external indicator.

CARE AND MAINTENANCE

- **Cleaning:** Use a soft, dry cloth to wipe the screen and enclosure. Do not use water, solvents, or abrasive materials.
- **Battery Caution:** Do not puncture, crush, or expose the Li-Po battery to heat or fire. Dispose of old batteries in accordance with local regulations.
- **Storage:** Store the PicoBoy in a cool, dry place away from direct sunlight. Keep the SD card slot and USB-C port free from dust.

SAFETY PRECAUTIONS

Read these precautions carefully before use.

Caution: Do not expose the device to water, moisture, or extreme temperatures.

Caution: Do not attempt to disassemble or modify the device other than via the intended USB reprogramming interface.

Caution: Use only 0.3V - 8V USB source to charge. Do not exceed the voltage rating and do not use damaged cables.

Caution: Do not short-circuit the battery terminals. Do not expose the Li-Po battery to physical damage.

Caution: If the device becomes unusually hot or emits a smell, power off immediately and disconnect from any power source.

TECHNICAL SPECIFICATIONS

The following section serves as a concise technical reference for the PicoBoy hardware and operating characteristics.

Parameter	Specification
Product Name	PicoBoy handheld gaming console
Main Controller	Raspberry Pi Pico W, RP2040 dual-core ARM Cortex-M0+
CPU Clock	Up to 133 MHz
On-Chip RAM	264 KB SRAM
Display Module	2.4-inch ILI9341 TFT LCD
Display Resolution	240 × 320 pixels
Display Interface	SPI
User Inputs	D-Pad, A, B, START, SELECT
Number of Buttons	8 tactile switches total
Motion Sensor	LSM6DSL 6-axis IMU
IMU Interface	I2C
Audio Output Method	PWM audio from RP2040
Audio Amplifier	LM386
Speaker	Mono speaker
Battery Type	3.7V rechargeable Li-Po
Battery Capacity	2000 mAh
Charging Port	USB Type-C
Firmware Update Port	Micro-USB
Bootloader Method	BOOTSEL + USB mass-storage flashing (.uf2)
Expansion	MicroSD slot reserved for future use
Enclosure Material	3D-printed PLA
Display Mode 1	ORIGINAL: 20 MHz SPI / 60 FPS
Display Mode 2	QUIET: 10 MHz SPI / 30 FPS
Development Environment	Raspberry Pi Pico SDK v2.2.0
Build System	CMake / Ninja

Main Schematic

Figure 14 shows the main system schematic for the PicoBoy hardware.

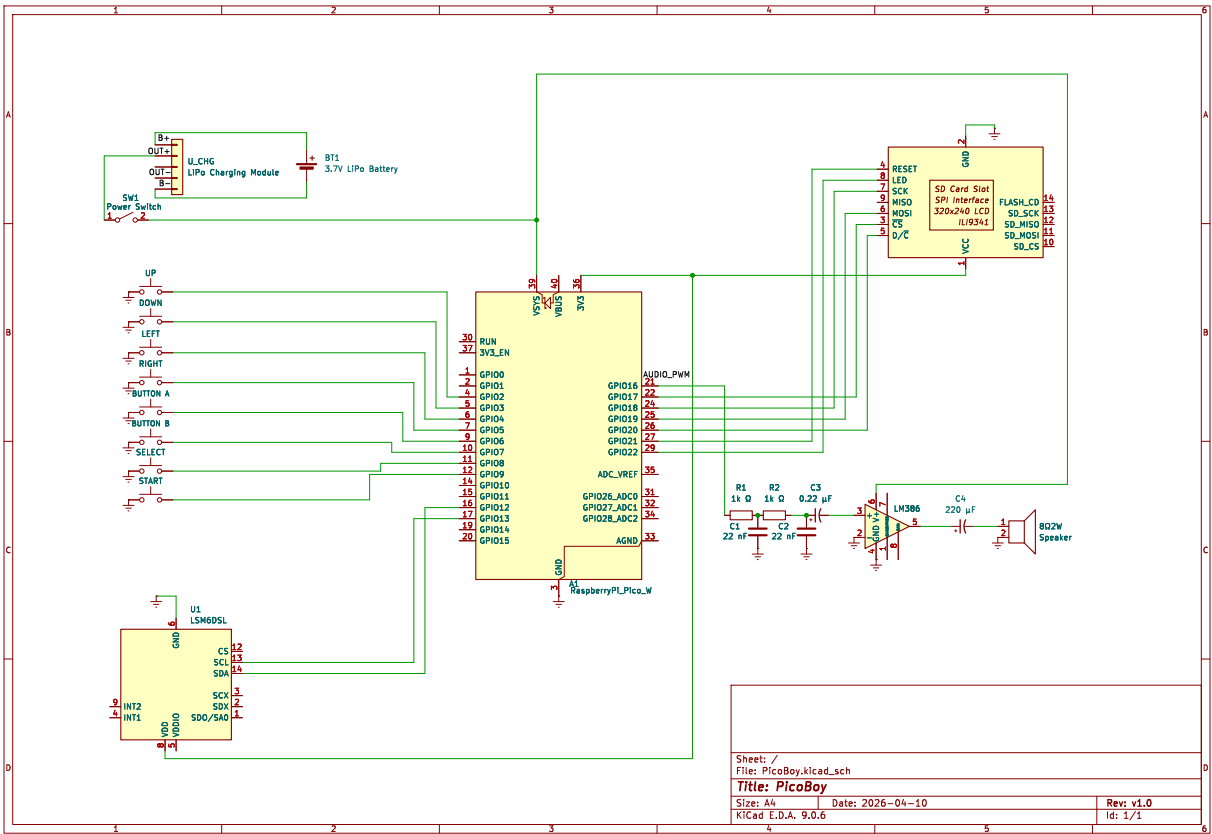


Figure 14: Main PicoBoy schematic